

EXPERIMENT No. 10

Date:

TO VERIFY THE OHM'S LAW AND DETERMINATION OF RESISTANCE OF GIVEN WIRE.

APPARATUS REQUIRED:

1. A meter wire
2. Ammeter
3. Voltmeter
4. Rheostat
5. Power supply
6. Key
7. Connecting wires

THEORY:

Ohm's law states that at constant physical conditions (temperature, length, etc) and small potential differences, the current flowing through a conductor is directly proportional to the potential difference applied across it. Mathematically,

$$I \propto V$$

Conversely, $V \propto I$

or, $V = IR$ (1),

where R is proportionality constant known as the electrical resistance of the conductor. Equation (1) is in the form $y = mx$, which is an equation of straight line passing through origin with slope m. Hence if we plot a graph of V versus I for a conductor wire, the result is a straight line passing through the origin. Then, Ohm's law is said to be verified. The slope of the straight line gives the value of resistance (R) of the wire of a particular length.

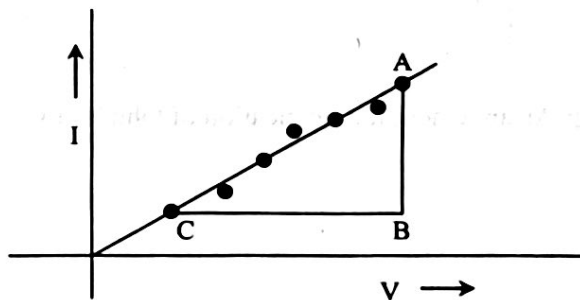


Fig.: I ~ V Graph

from graph

$$\text{Resistance, } R = \frac{1}{\text{slope}} = \frac{1}{\frac{AB}{BC}}$$

$$\therefore R = \frac{BC}{AB}$$

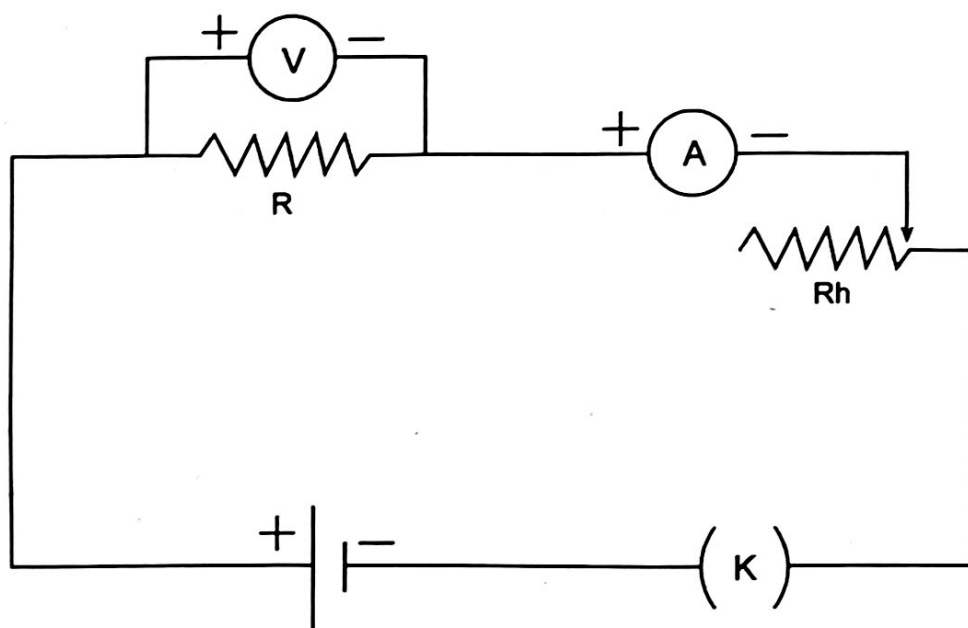


Fig: Arrangement for verification of Ohm's law

OBSERVATIONS:Least count of voltmeter = 0.1 VLeast count of ammeter = 0.1 ALeast count of meter scale = 0.1 cm = 0.001 m**Observation Table:**

S.N.	Length of the wire L (cm)	Ammeter Reading (A)	Voltmeter Reading (V)	$R = V/I$ (Ω)	Mean R (Ω)	R/L (Ω/cm)
	20	0.1	0.1	1	1	0.05
		0.2	0.2	1		
		0.3	0.3	1		
		0.4	0.4	1		
		0.5	0.5	1		
	40	0.1	0.2	2	2	0.05
		0.2	0.4	2		
		0.3	0.6	2		
		0.4	0.8	2		
		0.5	1.0	2		
	60	0.1	0.3	3	3	0.05
		0.2	0.6	3		
		0.3	0.9	3		
		0.4	1.2	3		
		0.5	1.5	3		

CALCULATION:

From the graph of I versus V,

For length 20 cm, slope $m = \frac{1}{2}$ $\therefore R = 1 \Omega$ For length 40 cm, slope $m = \frac{1}{2}$ $\therefore R = 2 \Omega$ For length 60 cm, slope $m = \frac{1}{3}$ $\therefore R = 3 \Omega$ **RESULT:**

Hence, the ohm's law is verified. The resistance of given wire is founded to be 1Ω at 20 cm, 2Ω at 40 cm and 3Ω at 60 cm.

CONCLUSION:

From the experiment, ohm's law is verified and resistance is found to be 1, 2, 3 at 20, 40 and 60 cm respectively

PRECAUTIONS:

1. The connection should be checked properly
2. The current should not be passed for long time. and should not be very high.
3. The ~~for~~ length of wire should be measured accurately.
4. Zero error of voltmeter and ammeter should be adjusted.
5. Reading should be taken carefully.

VIVA QUESTIONS:

1. Define resistance and specific resistance (Resistivity).
2. How does resistance vary with temperature?
3. Define potential and potential difference?
4. What are Ohmic and non-ohmic resistances?
5. If the radius of copper wire is doubled, what will be its resistance and specific resistance?
6. Define electric current.
7. Define conductivity. State its unit.
8. What is Shunt?
9. What is an ammeter? What is a voltmeter? How do they differ from a galvanometer?
10. How to convert galvanometer to ammeter?

1) The property of a substance which oppose the ~~to~~ flow of electric current is called resistance.

Specific resistance of substance is defined as the resistance across any opposite face of cube of unit side (1m) of that substance.

2) The resistance is directly proportional to its temperature.

3) Potential refers to the potential energy per unit charge at a specific point in an electric field.

Potential difference is the difference in electric potential between two points, representing the work needed to move a unit charge between ~~for~~ them.

4) Ohmic resistance is the type of resistance in which current is directly proportional to voltage, following ohm's law. ~~Non o~~

- Non ohmic resistance is the type of resistance in which current is not directly proportional to voltage, deviating from ohm's law.

5) If the radius of copper wire is doubled, its resistance will become one-fourth of its original value, while its specific resistance will remain unchanged.

6) Electric current is the flow of electric charge per unit time through a conductor or circuit, measured in Ampere (A).

7) Conductivity is defined as the property of a substance which allows the electric current to pass through it. Its SI unit is $\Omega^{-1} \text{m}^{-1}$.

8) A shunt is a low-resistance device connected in parallel to a circuit or component to direct excess current, commonly used in ammeter to measure high currents.

9) An ammeter is an instrument used to measure current passing through a conductor. A volt-meter is an instrument used to measure the potential difference across it.

A galvanometer is different from ammeter and voltmeter because a galvanometer is different from detects small electric currents whereas an ammeter measures current with low resistance in series, and a voltmeter measures voltage with high resistance in parallel.

10) A galvanometer can be converted into an ammeter by connecting a shunt resistor in parallel with it. This shunt resistor allows a portion of current to by pass the galvanometer.

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